



Examination Paper

Examination Session:

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Exam Code:

PSYC1062-WE01

PSYC1062 INTRODUCTION TO PSYCHOLOGICAL RESEARCH

Release Date/Time:	14:00 10 May 2022
Latest Submission Date/Time:	14:00 11 May 2022
Format of Exam:	Restricted Window Examination
Duration:	2 hours
Word/Page Limit:	N/A
Additional Material Provided:	None
Expected Form(s) of Submission:	Section B: Word Document using answer template provided. Your uploaded file should be named with your anonymous marking code ('Z number') and the Exam Code e.g., Z0123456 PSYC1062-WE01.
Submission Method(s):	Learn Ultra for Section A; Turnitin for Section B

Instructions to Candidates:

Answer ALL Questions in Section A.

Answer ALL Questions in Section B.

Section B

Complete this section using the answer template provided.

Question 1:

A developmental psychologist conducted a study to investigate 18-month-old children's behaviour in an isolation situation. They filmed 14 children and their mothers playing together. At some point, the mother was asked to hide from their child until the child reacted and called for the mother's presence. The mother then returned and started playing again with the child. The psychologist measured the reaction time (RT) for each child, which is the time between the moment the mother hides and the time the child calls for her presence. RTs are measured in seconds. The psychologist then reproduced the same scenario with the same mother/child pair, the second round being 20 minutes after the first round.

Data are presented in the table below:

Table 1. Child reaction time (in seconds)

Child	First Round	Second Round
1	30	25
2	25	20
3	15	12
4	25	30
5	38	30
6	10	10
7	14	10
8	20	15
9	18	15
10	8	8
11	32	27
12	18	12
13	25	22
14	15	18

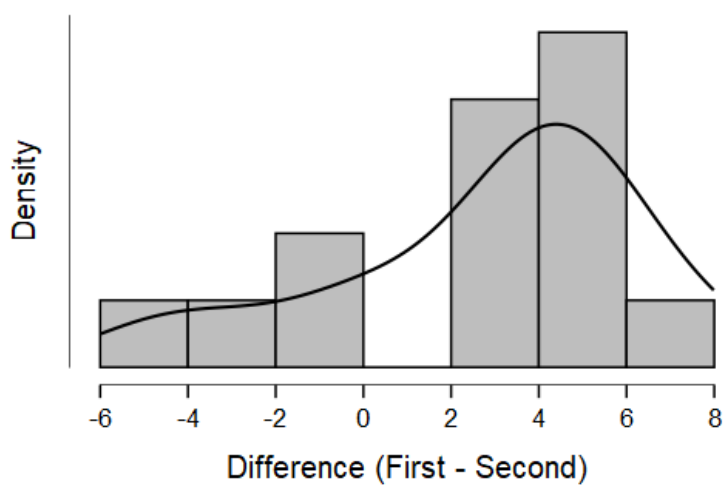
- a. What is the design of the study? **(1 mark)**
- b. Identify the independent and the dependent variables. **(2 marks)**
- c. Should the psychologist run a two-tail or a one-tail test? Why? **(2 marks)**
- d. What is the null hypothesis? And the alternative hypothesis? **(2 marks)**
- e. The psychologist was not sure about which test to use so they ran two different analyses, the results of which are reported below. Select the most appropriate and justify why. **(6 marks)**
- f. Summarize the findings and give a conclusion. **(7 marks)**

Data familiarisation

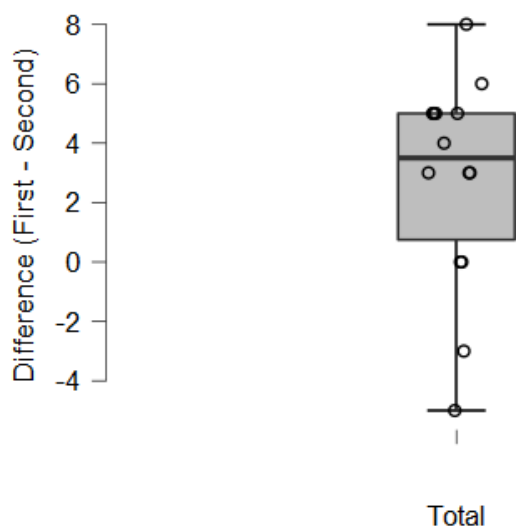
Descriptives

	N	Mean	SD	SE
First Round	14	20.93	8.63	2.31
Second Round	14	18.14	7.62	2.04

Distribution Plots



Boxplots



Results

Measure 1	Measure 2	Test	Statistic	df	p	Effect Size
First Round	Second Round	Student	2.89	13	0.012	0.77
		Wilcoxon	67.5		0.026	0.73

Question 2:

A developmental psychologist was interested in the relationship between children's sociability and their musical aptitude. He rated 12 children for sociability using a scale of 0-14 (where the higher the number the greater he perceived their sociability) and also assessed their musical aptitude using a battery of non-standardised tests (where the higher the score the greater the child's musical aptitude). The scores are shown in Table 2. The psychologist predicted that there would be a significant negative relationship between the sociability ratings and the musical aptitude test scores.

Table 2. Children's sociability ratings (on a scale of 0-14) and their musical aptitude test scores (on a scale of 0-100).

Child	Sociability rating (0-14)	Music aptitude test score (0-100)
1	1	97
2	3	80
3	4	67
4	5	60
5	6	55
6	7	40
7	8	25
8	10	25
9	11	20
10	12	25
11	13	40
12	14	85

- a. Draw an appropriate diagram to illustrate the relationship between 'Sociability rating' and 'Musical aptitude score'. **(6 marks)**
- b. The developmental psychologist was not certain which statistical analysis would be appropriate to ascertain whether the data supported his prediction. He carried out two different tests using JASP. The output from these analyses is shown below.

Spearman's Correlations

			Spearman's rho	p
Sociability Rating	-	Music Aptitude test score	-0.504	0.094

Pearson's Correlations

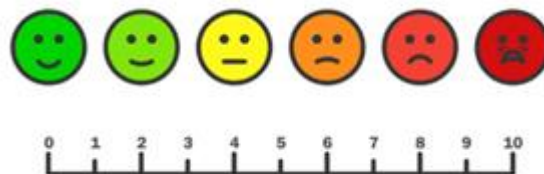
			Pearson's r	p
Sociability Rating	-	Music Aptitude test score	-0.536	0.073

Which is the most appropriate output? Give reasons for your choice. **(4 marks)**

- c. What conclusion can you make about the relationship between 'Sociability' and 'Musical aptitude'? Report this as you would in the Results section of a practical report. **(6 marks)**
- d. What features of this data set are not revealed by the statistical analysis? **(4 marks)**

Question 3:

A group of researchers is interested in working-related stress in academics, and more specifically under lockdown conditions. They wanted to compare stress level between Academics in the UK experiencing work under full lockdown and Academics in Sweden who did not experience lockdown conditions. They asked 15 academics from a UK university and 15 academics from a Swedish university to rate their stress level on a scale from 0 to 10 as shown below:



The researchers' assumptions are that there will be a difference in stress level between UK and Swedish academics, and that UK academics should report higher stress levels than Swedish academics.

The results are reported in the table below:

Table 3. Stress level rates for academics in UK and Sweden

UK	Sweden
7	4
6	3
4	1
8	3
7	5
5	3
4	1
6	0
5	2
4	4
8	1
10	1
5	6

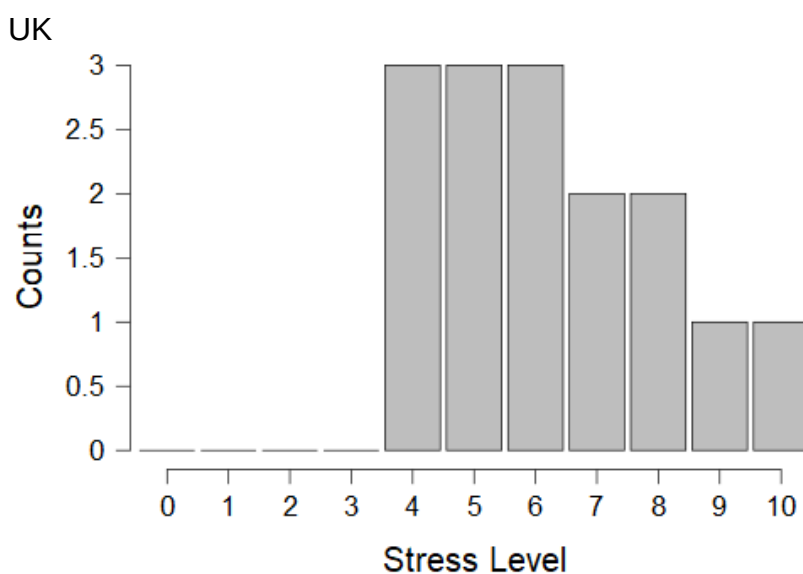
6	1
9	5

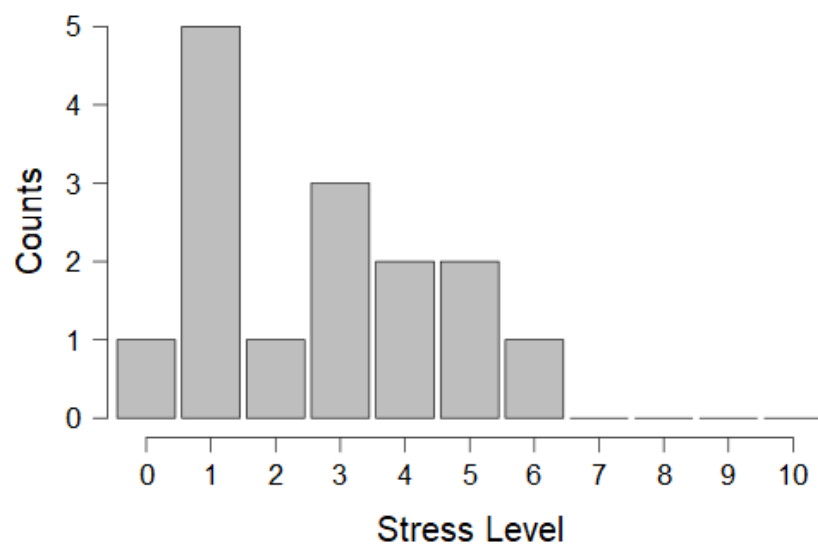
- What is the design of the study? **(1 mark)**
- Identify the independent and the dependent variables. **(2 marks)**
- Should the researchers run a two-tail or a one-tail test? Why? **(2 marks)**
- What is the null hypothesis? And the alternative hypothesis? **(2 marks)**
- In order to examine the data before analysing it, the psychologist drew histograms (shown below). Describe the data shown in these plots. **(2 marks)**
- The researchers perform a statistical test to test for their hypothesis. Results are presented below. What test did they perform and why? **(4 marks)**
- Summarise the findings and give a conclusion. **(7 marks)**

Data familiarisation

Group Descriptives

	Group	N	Mean	SD	SE
Stress Level	Sweden	15	2.67	1.84	0.47
	UK	15	6.27	1.87	0.48



**Results**

Variable	Statistic	p	Effect Size
Stress Level	19.500	< .001	0.827

END